

SCR Measure — Subsea

The deepwater version, with riser pressure, choke-line pressure and riser margin

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

Well Control

Subsea BOP

Riser margin

A4 PDF

1. What It Is

The Subsea SCR Measure screen is the deepwater counterpart of the Surface one. It records slow-rate pump pressures the same way, but adds the extra measurements and calculations that a **subsea BOP** needs. It is reached from Operations › Drilling › Well Control › SCR Measurement › Subsea.

Accent colour	Teal, distinct from the cyan used for Surface
Output	A4 field sheet PDF through the Share menu
Saving	Three named slots on the device
Also inside	A subsea BOP training simulator on a second top tab

2. What Makes Subsea Different

A subsea BOP sits on the seabed and is joined to the rig by a **marine riser**. That creates three extra considerations:

- The **choke line** runs from the seabed to the surface, so its friction pressure must be measured and tracked separately from the riser pressure.
- The riser holds a column of fluid that can differ in density from the mud in the drill string, giving a pressure imbalance called the **riser margin**.
- Water depth and riser length affect both the hydrostatic pressures and how SCR readings should be interpreted.

Main structural difference

The SCR table has **three** columns per row — speed, riser pressure and choke-line pressure — instead of two. Section 4 is entirely subsea-specific and has no equivalent in the Surface screen.

3. The Five Sections

#	Section	Contents
1	Well / Job Info	Well, field, rig, operator, date and engineer
2	Pump Specification	Liner, stroke, rating, displacement (automatic), efficiency and maximum pressure
3	SCR Pressure at Speed	Six rows of speed, riser pressure and choke-line pressure
4	Subsea Parameters	Water depth, riser length, seawater density (default 8.6 ppg), riser mud weight, choke-line friction, kill-line friction and active mud weight
5	Summary	Primary SCR pressure, secondary SCR pressure and Riser Margin pressure

4. How to Use It

1. Fill in the well and job information.
2. Enter the pump specification. The liner size can be in inches or millimetres.
3. Run the pump at each target speed for at least three minutes and record **both** the riser pressure and the choke-line pressure on each row.
4. Enter the subsea parameters.
5. Read the summary, which updates as you type.
6. Save to one of three slots, Open a saved one, or Export the A4 PDF.

5. The Calculations

Result	How it is worked out
Pump displacement	Same triplex formula as Surface: $0.000243 \times \text{liner}^2 \times \text{stroke length}$
Riser margin pressure	$(\text{riser mud weight} - \text{seawater density}) \times 0.052 \times \text{riser length}$. The constant 0.052 turns pounds-per-gallon $\times$ feet into psi
Primary / secondary SCR pressure	Taken from the riser pressure column: the lowest and next-lowest recorded values

Why riser margin matters

The riser margin should stay positive. A zero or negative value points to a risk of riser unloading, where gas or formation fluid pushes the riser fluid out.

The riser margin recalculates live as any of its five inputs change.

6. The Simulator Tab

A second top tab, **Simulator**, holds a subsea training simulator with three inner tabs:

- **BOP Control** — the stack's annular, pipe ram and blind shear ram, each with open, block or close status lights.
- **Mud Pumps** — start and stop, speed control and a live gauge, with choke- and kill-line read-outs.
- **Driller's Console** — weight on bit, rotary speed, standpipe pressure and casing pressure indicators.

7. The PDF

A single A4 page: a teal header with the well, date and engineer, a well-identification grid, a pump-specification table, the **SCR table and subsea parameters side by side**, a results block with primary SCR, secondary SCR and riser margin, then notes and a signature block.

8. Surface vs Subsea at a Glance

Feature	Surface	Subsea
Accent colour	Cyan	Teal
SCR table columns	Speed + standpipe pressure	Speed + riser pressure + choke-line pressure
Extra parameters	None	Water depth, riser length, densities, friction
Extra calculation	None	Riser margin pressure
PDF layout	SCR table at full width	SCR table beside the subsea parameters
Save slots	Three	Three